Chapter 11 — Special Theory of Relativity

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Special relativity applies to: (a) rotating only (b) gravitational only (c) accelerating only (d) inertial frames

2. Speed of light is: (a) greater if source moves (b) zero (c) less if observer moves (d) same for all observers

3. Moving clocks run: (a) same (b) backwards (c) slow (d) fast

4. Length contraction occurs: (a) all directions (b) at rest (c) perpendicular (d) along motion

5. Relativistic mass: (a) m_0c^2 (b) $m_0/\sqrt{(1-v^2/c^2)}$ (c) $m_0\sqrt{(1-v^2/c^2)}$ (d) m_0v/c

6. Mass–energy relation: (a) $E = \frac{1}{2}mc^2$ (b) $E = m/c^2$ (c) $E = mc$ (d) $E = mc^2$

7. Relativistic K.E.: (a) $(m-m_0)c^2$ (b) mc^2 (c) $\frac{1}{2}mv^2$ (d) m_0c^2

8. Material body cannot reach: (a) escape speed (b) orbital speed (c) sound speed (d) c

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is an accelerating car a non-inertial frame?

2. Why is proper time always the shorter time interval?

3. Why does length contraction occur only along the direction of motion?

4. Why are relativistic effects not noticed in daily life?

5. Why can a material body not reach the speed of light?

6. How does $E = mc^2$ explain large energy in nuclear reactions?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) State Einstein’s postulates and explain time dilation and length contraction. **(4)**

(b) Explain relativistic mass variation and why c is the ultimate speed limit. **(4)**

Q.4 (a) Explain Einstein’s mass–energy relation and rest energy. **(4)**

(b) Give examples where special relativity has been verified. **(4)**

studysapiens.com